

Music as a Tool for Social Bonding: A Survey

By

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Abstract

The Republic III (Plato, 360 BC/2006), Plato attributes to music a powerful influence on the personality of its audience. This philosophical perspective, that music may positively shape how we act, feel and behave, still holds in our present day and time and is manifested transculturally in both Western and non-Western traditions. It further induces that music, as a universal language, profoundly impacts human emotional and social development throughout the lifespan. From early infancy, individuals are attuned to musical structures, which fosters emotional awareness and social connections. However, in order to truly understand such processes, their underlying mechanisms and the benefits that one can derive from them, there is need to establish a science that can systematically link between musical and social skill. This explains that such theories of musicality must focus mainly on the value of music for specific adaptive contexts such as mate selection, parental care, coalition signaling, and group cohesion. Synthesizing and extending this, it is argued that social bonding is an overarching function that unifies all of these theories, and that musicality enabled social bonding at larger scales than grooming and other bonding mechanisms available in ancestral primate societies. This paper examines the role of music in developing emotional intelligence, empathy, and social bonds. To achieve it, Functionality Theory and Qualitative Research Design are employed as guide. This paper will conclude by recommending that considering music's unique capacity to enhance emotional and social growth, practical applications to improve resilience, inclusivity, and community cohesion should be obtained pedagogical purposes.

Keywords: Music, Tool, Social Bonding, Pedagogy, Development

Introduction

Music according to Kakembo Aisha Anne (2024) is a universal language present in all human cultures, used to express thoughts, sublimate emotions, and enhance social cohesion. From a developmental point of view, it accompanies a human being permanently across their entire life span. Infants have been shown to perceive and classify musical structures from birth and are able to extract properties of musical-like structures from languages spoken around them. Furthermore, they prefer to listen to vocal music compared to speech and have "protomusical" interactions infused with positive valence. This versatility of music throughout the lifespan and across different cultures suggests that it may interface with brain structures and cognitive processes co-opted in evolution and conserved across humanity to enhance emotional communication and social interaction (Nosirov, 2021, in Kakembo, 2024).

The experience of playing or listening to music touches a variety of psychological aspects, including emotion and behavior. It may be for these reasons that the use of music experience for developing emotional and social abilities is the object of growing interest both in the field of pedagogy, with the aim of educating to the full the growing citizen, and in the field of

psychomotricity and psychology, as the facilitator of experiences that affect growth and health. Viewing different programmatic documents, one can observe the incursion of music and the arts in general in the educational sphere (Kakembo, 2024). In the Italian school documents, a clear reference to the importance of music is recorded. Let's analyze in brief how the issue of music as an emotional learning tool is introduced and a few data related to international guidelines. It is interesting to note that this document was designed not only to illustrate the essential knowledge that, the type of school referred to as Kindergarten for children from zero to six years, must be included in the educational offer by the Functional Area of Music and in particular to use music as an educational and formative vehicle by giving young learners an emotional world capable of expressing itself thanks to the use of technical means, but also to provide in a general way ideas and suggestions in relation to the involvement of children from 0 to 6 years in relation to the level of understanding (Kakembo, 2024).

It is against this that Savage, Loui, Tarr, Schachner, Glowacki, and Mithen (2021) specifically noted that Darwin (1871) famously considered music to be a puzzle for evolutionary theory. Music according to them is universal across human cultures, yet its function seems mysterious, since “neither the enjoyment nor the capacity of producing musical notes are faculties of the least use to man in reference to his daily habits of life...”. Darwin went on to speculate that music first evolved “for the sake of charming the opposite sex”, after which language “derived from previously developed musical powers”.

More so, Savage et al (2021) noted that since Darwin there has been no shortage of hypotheses about why and how music evolved. The null hypothesis is that music is an evolutionarily “useless” by-product of other evolved capacities, with no adaptive function and involving no direct selection for musical capacities (Pinker 1997, in Savage et al, 2021). Others hypothesize according to Savage et al that musicality evolved for specific adaptive purposes, including signaling mate quality (Miller 2000), advertising male coalitions (Hagen & Bryant 2003; Merker 2000), or soothing infants. Many authors according to them have discussed the evolutionary value of music in facilitating group cohesion, sometimes suggesting that music may have arisen via group selection (especially Brown 2000a). Although such proposals according to them succeed in explaining some properties (or genres) of music, we argue that no single account succeeds as a general explanatory framework for the evolution of human musicality.

My purpose in this article is to synthesize and extend previous proposals into a new, parsimonious framework that can explain and predict many aspects of human music-making which provides a framework for understanding the biological and cultural evolution of music, taking this argument as foundational. I provided a detailed cross-disciplinary review of the evidence for specific mechanisms by which music functions to enhance social bonding, and consider how some of the mechanisms underlying musicality may have coevolved with music.

Social bonding as a unifying function in the evolution of musicality

The Music and Social Bonding (MSB) hypothesis Fitch (2015) as noted in Savage et al (2024) posits that core biological components of human musicality evolved as mechanisms supporting social bonding. Musicality relies on multiple neurocognitive components, which likely evolved at different times and for different reasons: musicality is more a cognitive toolkit than a single

tool. Most of the tools in this musical toolkit function to facilitate social bonding, but some may also be used for non-social purposes such as individual mood regulation.

This according to Savage et al explains that Coherent and harmonious merging of sounds and movements during group activity leads to positive feelings of prediction, fulfillment of expectation, and mutual accomplishment. These, through activation of the dopaminergic reward system and other pathways, have affiliative emotional and rewarding effects immediately and also long after music-making ceases. Crucially, the resulting strengthened social bonds are operative over multiple types and sizes of groups, ranging from dyads (e.g., parent and infant, potential mates) to bands of small coalitions and large groups of unrelated individuals. Social bonding through music thus produces its ultimate evolutionary dividends in multiple complementary ways, including a larger group of potential allies, increased child rearing success, increased mating success, and better-functioning coalitions (Salvage et al, 2024).

Music and Social Identity

A final potential design feature of culturally-transmitted group music concerns its role in flexibly and hierarchically indicating kinship and group identity (Stokes 1994; Turino 2008 in Salvage et al, 2024). Because songs are variable, complex and memorable, two people knowing the same song likely acquired this knowledge via social learning -- and thus are likely to share a common socio-cultural history. Thus, shared knowledge of musical repertoire provides information about shared socio-cultural background (Soley & Spelke 2016; Schachner et al. 2018). Musicality may have coevolved in support of this social bonding function: Cultural innovations created a wide variety of musical styles and features, and musical knowledge became a cue to social history and cultural group membership. This created selective feedback favoring individuals who tended to perceive music as a cue to group membership, as they would have more accurate ideas about others' social group membership. This hypothesized combination of cultural and biological evolution would lead to an evolved bias to use music as a cue to guide and facilitate social interactions, consistent with findings that shared musical knowledge serves as a social cue from early in childhood through adulthood (Soley & Spelke, 2024).

Synchronized and harmonized group performances according to Soley and Spelke help cement group identity, and eventually allow skilled participation in ritualized performances to serve as a hard-to-fake indicator of group membership. Furthermore, the existence of diverse pieces and sub-styles allows subgroups to express their uniqueness within a broader shared musical repertoire or style. Such expressions of identity at multiple hierarchical levels are useful because human biological and cultural evolution has been characterized by increasing complexity of social structure, as exemplified by the large-scale nation-states characteristic of modern human societies (Turchin et al. 2018). Thus, group musical performance - including dance - facilitates lasting, culturally evolving indicators of group identity and bonds - akin to passwords or shibboleths (Feekees 1982, Fitch 2004 in Belfi & Loui, 2020) – that extend beyond individual recognition and memory, aiding intercultural marriage and trade.

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Psyche Louib, Bronwyn Tarr, Adena Schachner, Luke Glowackie, Steven Mithen and Tecumseh Fitch (2021) posits that core biological components of human musicality evolved as mechanisms supporting social bonding. Musicality relies on multiple neurocognitive components, which likely evolved at different times and for different reasons: musicality is more a cognitive toolkit than a single tool. Most of the tools in this musical toolkit function to facilitate social bonding, but some may also be used for non-social purposes such as individual mood regulation. We avoid arguing for one specific single adaptive function for music (e.g., coalition advertisement, courtship, or infant mood regulation) because we think it unlikely that a single “main” evolutionary function for complex, multi-component abilities such as language or music exists (Patrick et al, 2021).

Imagine asking the parallel question “what is vision for?” and coming up with a hypothesis set including “spotting predators,” “judging mate quality,” “finding food,” and “avoiding obstacles.” It seems clear that these are all functions of vision, and all provide potential causal explanations for adaptive improvements in vision during evolution. But the desire to identify ONE function as primary seems misguided. A better approach is mechanistic: we ask “what are lenses for?” and answer in engineering terms: lenses are for focusing an image on the retina, to enable accurate visual perception. Whether the image is of a predator, mate, or food is not critical, because of improved visual resolution will aid them all. Turning to music, “social bonding” provides an umbrella explanation analogous to “vision is for seeing (Patrick et al, 2021)”.

Particular design features of music (singing discrete pitches, generating an isochronous beat, and use of repetitive patterns based on small-integer ratios) function mechanistically to enhance predictability, aiding synchronization and harmonization when multiple people sing, dance, and play instruments together. Coherent and harmonious merging of sounds and movements during group activity leads to positive feelings of prediction, fulfillment of expectation, and mutual accomplishment. These, through activation of the dopaminergic reward system and other pathways according to Patrick et al have affiliative emotional and rewarding effects immediately and also long after music-making ceases. Crucially, the resulting strengthened social bonds are operative over multiple types and sizes of groups, ranging from dyads (e.g., parent and infant, potential mates) to bands of small coalitions and large groups of unrelated individuals. Social bonding through music thus produces its ultimate evolutionary dividends in multiple complementary ways, including a larger group of potential allies, increased child rearing success, increased mating success, and better-functioning coalitions (Patrick et al, 2021).

Functionality Theory

Every discipline is grounded on its own set of theories which develop over a particular point of disciplinary history. Unfolding of the discipline can be better understood in terms of its (i) theoretical rigor and (ii) methodological orientation. It's the theory that provides the broad framework or orientation for interpretation of facts and the methodology provides specific rules, the logical guidelines for collection and analysis of the data in this regard. Theory provides the template of ideas to think, methodology provides the techniques for collection of ideas so that they can be logically connected to one another in form of a theoretical framework.

Functionalism Theory according to Gloria Gómez-Diago (2019) was propounded by Emile Durkheim in 1890s. He talked about four basic "instrumental needs" (economics, social control, education, and political organization), that require institutional devices to get fulfilled. While Radcliffe-Brown focused on social structure rather than biological needs. He considered society as a system. He looked at institutions as orderly sets of relationships whose function is to maintain the society as a system.

Functionalism according to Gloria is a theoretical perspective that focuses on the functions performed in society by social structures such as institutions, hierarchies, and norms. Within this theory, function refers to the extent to which a given activity promotes or interferes with the maintenance of a system. functionalism which has been considered one of the prominent schools of thoughts in order to understand various aspects of culture and society. Functionalism arose as a reaction to evolutionism and diffusionism in early twentieth century. Functionalism looks for the function or part that is played by several aspects of culture in order to maintain a social system. It is a framework that considers society as a system whose parts work together to promote solidarity and stability. This approach of theoretical orientation looks at both social structure and social function. It describes the inter-relationship between several parts of any society. These parts or the constituent elements of a society could be named as norms, traditions, customs, institutions like economy, kinship, religion etc (Gloria, 2019).

This study adopts Functionalism Theory because it explains how social order is maintained through solidarity, how breakdowns lead to anomie, and why religion and morality are central to social life. His theory laid the foundation for functionalism and modern sociology.

Methodology

Methodology in science refers to the diverse principles, procedures, and practices that govern empirical research. It is useful to distinguish five major components to convey the scope of the topics and to organize the subject matter. This paper adopts Qualitative approach to research because it concerned with the interrogation of subjective assessment of attitudes, opinions and behaviour. It further makes use of secondary data, such as reputable journals, textbooks, articles and lecture notes. And primary data such as the writer's basic idea of the population area.

Benefits of Social Bonding

We hypothesize that musicality increased the number of "simple" relationships (e.g. "friends"), and increased the quality (depth and complexity) of existing relationships. The opportunity for many individuals to participate productively in social interaction through proto-musical behaviors facilitates an efficient bonding mechanism for groups of varying sizes, thereby conferring associated benefits. However, we must consider the nature of the subsidiary relationships and social structures in which they operate. Many vertebrate species live in large groups (e.g., fish schools, bird flocks, ungulate herds), but do not exhibit strong social bonds with more than a small number of individuals, and/or the relationships are undifferentiated. Indeed, the "number of differentiated relationships" (Bergman and Beehner, 2015) can vary independently from raw group size. For example, a monogamous pair with bi-parental care involves two differentiated relationships (sexual mate, and caregiving partner) or even three (adding joint territory defense), a situation typical in many birds. The social bonding design

features we have identified can operate at multiple levels simultaneously, in the same way that a couple dancing at a party can intensify their own relationship, and their relationship with the broader social group.

The Impact of Music on Emotional Development

One of the most studied areas in the domain of personality and social interaction according to Kakembo (2024) is emotional development. Emotional changes are more common in infancy and adolescence than in any other period of life. Considering this, it is therefore relevant to examine how music influences the emotional development of people in different age groups. Psychological studies have established a strong link between music and human emotions. Music's nuanced arpeggios according to Kakembo might evoke nostalgia; a grandiose symphony might uplift our mood and make us feel happier. Music can also explore dark harmonic territories and elicit feelings of sadness and anxiety. This is why the emotional domain is one of the most examined areas of music psychology. The proposal informed several theories of 'emotional intelligence' associated with music exposure; theories suggesting that engagement with music can help develop emotional awareness through the occurrence of various emotional states during active music listening or music-making (Kakembo, 2024).

A study of adults found correlations between music preference and well-being, with people who identify with popular musical genres such as pop/rock, dance, and easy listening reporting significantly higher subjective well-being in terms of life satisfaction, self-esteem, and emotional listening mechanisms. This, according to Kakembo supports a significant negative correlation between people's preference for musical genres and subjective well-being, increasing the ability to cope with emotions and express them openly. In short, listening to music is a way for people to express their emotions. This is also the reason why people who do not express their emotions usually feel incomplete. Listening to music or engaging in music-making is also used informally to release or treat emotional disturbances. Rhythm favored emotional catharsis and made the emotional lives of his well-being more acceptable. This is why listening to beautiful music or participating in musical activities that produce positive effects on the human mind is justified, which can relieve anxiety, calm nerves, control emotions, and promote mental health (Kakembo, 2024).

Music And Social Development: Building Connections and Empathy

Listening to and playing music according to Kakembo (2024) is a social activity with deep emotional significance. There are numerous opportunities for group musical activities in a musical lifespan, such as attending concerts and other live performances, formal community or religious music groups, or more casual interaction and music-making with friends and family. Everybody has different musical experiences, and there is always something new to learn. Shared experiences, whether attending a basketball game, religious service, or music festival, help to unite people who enjoy the same music. These activities also provide a sense of community, and in most cases, the individual can adapt their feelings to the emotions of the group. This adaptability is a crucial social skill that takes time to develop and requires a great amount of self-awareness (Kakembo, 2024).

The role of music and other arts in encouraging feelings of empathy and compassion is

now recognized widely. Potential drivers to encourage tolerance and understanding are used in schools to promote an inclusive society. The discipline required to participate in a band or choir, for example, teaches teamwork and leadership skills. Musical and life lessons are learned in a choral setting that help the other members of the group appreciate diverse perspectives. Because of this, the connections made by participation in musical activities are not just with music; they are also with one's peers. Understanding music can encourage an appreciation of the emotive qualities of being human and an understanding of someone else's feelings. Students are working to better understand a piece of their research in order to better reflect what the composer intended. Active participation in music has the power to bring us together as a people. Four fifty-minute classes are more introspective than a group that plays in a band setting. You have the opportunity in music to play in a group or alone (Kakembo, 2024).

Conclusion

Social bonding has long been acknowledged as an important function of contemporary music, but its causal role in the evolution of music has often been dismissed as a naïve application of group selection theory. Recent advances in gene-culture coevolution theory allow us to provide a more nuanced model of music evolution that does not rely on group selection. Our argument has focused on social bonding as the primary factor shaping the evolution of human musicality. This MSB hypothesis provides a framework for understanding the past evolution of musicality, and a starting point for the future cultural evolution of new forms of music that harness the social power of music to bring people together. Music may not be a “universal language, but music's universal power to bring people together across barriers of language, age, gender, and culture sheds light on its biological and cultural origins, and provides humanity with a set of tools to create a more harmonious future - both literally and figuratively.

This explains that music profoundly influences emotional and social development, beginning in infancy and extending through adulthood. It is not only a means of personal expression but also a tool that fosters emotional resilience, empathy, and social skills. Musical engagement promotes a sense of belonging and cultural identity, while shared musical experiences cultivate empathy and understanding across diverse groups.

Recommendations

The following recommendations are therefore proposed:

1. Practical application by all stakeholders: considering music's unique capacity to enhance emotional and social growth, practical applications to improve resilience, inclusivity, and community cohesion should be obtained pedagogical purposes.

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